

industry employs almost 50,000 professionals today. "This demand will be at least double in the next three years. Science and engineering graduates having an aptitude for machine control will find space for a fulfilling professional career here," adds Wadhwa.

Money matters

Speaking of salary prospects, a fresh engineering passout may start at Rs 300,000-Rs 400,000 per annum. However, the scene is not so encouraging for diploma holders as their starting salary usually ranges from Rs 150,000 to Rs 200,000 per annum.

You could start at a junior level as part of a major project and grow to become a project leader in 10-12 years' time frame. In most cases, salaries are proportional to the cost of the projects. Even in this time of economic downturn, the automation industry is offering a salary in the range of Rs 500,000 to Rs 1 million per annum for professionals with five to seven years of experience.

Notably, the salary is on the higher side for design section. And if you grab an international opportunity, the minimum salary may be in the range of Rs 2 million per annum within just about five years.

Groom for automation

Now that you have received enough boosters about the opportunities to be grabbed in the automation field, it's time to explore the power circuit. The automation field is specifically suitable for engineering professionals with multidisciplinary as well as project management interests, such as project engineering. Exciting and technically satisfying careers can be pursued in the field of technical marketing, design engineering, project management, integration and servicing. And if you consider technology, I must say, wherever there is a need for process control to increase productivity, there is automation. It may be a human gene analysis laboratory or a locomotive workshop.

The main metric is grip on con-

trol. And as a professional you are expected to acquire the knowledge and skills needed to design practical control loop as per that metric. If you can achieve these, instrumentation and maintenance related solutions will be automatically on your fingertips. To make maximum use of the situation, you need to have expertise to survive in the global-quality working environment of automation.

Since the automation industry comprises fairly large and mature companies with highly technical processes, this reflects on the manpower needs as well. The skill requirements are vast and there is room for professionals for product design and development, sales, project engineering, product management, marketing and sales. Professionals who have a superior understanding of electronics, instrumentation, electrical, mechanical and robotics engineering are required among others.

Automation is a very technical field and therefore domain expertise and knowledge of processes is very important. Because of the technology-driven nature of this field, professionals must at all times keep themselves abreast of the latest improvements and technological upgradations that are taking place, instead of narrowly focusing on their departmental concerns.

In addition to technical skills, recruiters are looking for such competencies as interpersonal skills and drive for end results down the ranks. Don't get nervous. You are not expected to know everything but your learn-ability levels should be high as a lot of training would essentially take place on the job.

Next, let me share some basics of automation beyond programmable logic control (PLC) and SCADA.

How to control the control loop?

Keep in mind that automation is the use of control systems (such as numerical control, programmable logic control and other industrial control systems), along with other applica-

tions of information technology (such as computer-aided design and computer-aided manufacturing), to control industrial machinery and processes, thereby reducing the need for human intervention.

A simple process control loop consists of three elements: a measurement system, a controller and a final control element. With regards to industrialisation, automation is a step beyond mechanisation. While mechanisation provided human operators with machinery to assist them in the physical requirements of their work, automation greatly reduces the human sensory and mental requirements. It refers to a wide range of hardware and software products and protocols used to communicate between standard computer platforms (PC, Macintosh or workstation) and devices used in industrial automation applications such as controllers.

Before the advent of computers, controllers were usually single-loop proportional-integral-derivative (PID) controllers. These were used to execute PID control functions. These days, the controllers can do a lot more, however, 80 to 90 per cent of the controllers used in India are still PID controllers. Digital controllers do not have mechanical moving parts. Instead, these use processors to calculate the output based on the measured values. Since they do not have moving parts, they are not susceptible to wear and tear with time. However, digital controllers are not continuous.

Understand the controllers from both the theoretical and practical point of view. Forget the analogue versus digital controversy. Both types of controllers have their respective pros and cons. Analogue controllers are based on mechanical parts that cause changes to the process via the final control element. Again, like final control elements, these moving parts are subjected to wear and tear over time and that causes the response of the process to be somewhat different with time. But analogue controllers control